

# IMSC-2025-Functional equations of Mongolian Mathematical Olympiad

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## 1. Mongolian Mathematical Olympiad - 2024

Let  $\mathbb{N}$  denote the set of all natural numbers. Find all functions  $f: \mathbb{N} \rightarrow \mathbb{N}$  such that  $f(1) = 1$  and

$$f(f(n)) = 3f(n) - 2n, \quad \frac{n-1}{f(n)-n} \text{ is integer}$$

for all  $2 \leq n \in \mathbb{N}$ .

## 2. Mongolian Mathematical Olympiad - 2023

Let  $\mathbb{R}_{>0}$  denote the set of positive real numbers. Find all pairs  $f, g: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$  of functions such that

$$f(g(x)) = f(x)g(x), \quad f(x) = x(1 + g(x))$$

and such that the sequence  $x, g(x), g(g(x)), \dots$ , takes finitely many different values for all  $x \in \mathbb{R}_{>0}$ .

## 3. Mongolian Mathematical Olympiad - 2022

Let  $\mathbb{R}$  denote the set of all real numbers. Find all pairs of functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  and  $h: \mathbb{R}^2 \rightarrow \mathbb{R}$  such that

$$f(x + y - z)^2 = f(xy) + h(x + y + z, xy + yz + zx)$$

for all  $x, y, z \in \mathbb{R}$ .

## 4. Mongolian TST - 2020

Let  $\mathbb{R}$  denote the set of all real numbers. Find all functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(f(x) + y) = f(f(x) - y) + 4f(x)y$$

for all  $x, y \in \mathbb{R}$ .

## 5. Mongolian Mathematical Olympiad - 2023

Let  $\mathbb{R}_{>0}$  denote the set of positive real numbers. Find all functions  $f: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$  such that

$$f(x)f(y + f(x)) = f(1 + xy)$$

for all  $x, y \in \mathbb{R}_{>0}$ .

**Problem - 1.** Let  $\mathbb{N}$  denote the set of all natural numbers. Find all functions  $f: \mathbb{N} \rightarrow \mathbb{N}$  such that  $f(1) = 1$  and

$$f(f(n)) = 3f(n) - 2n, \quad \frac{n-1}{f(n)-n} \text{ is integer}$$

for all  $2 \leq n \in \mathbb{N}$ .

**Answer:**  $f(n) = 2n - 1$

**Solution:**

It is easy to check that the above function is a solution, so let us prove that there are no other solutions. If  $f(n) = f(m)$ , we have  $2n = f(f(n)) - 3f(n) = f(f(m)) - 3f(m) = 2m$ , thus  $f$  is injective.

We have  $2 \leq n$  then  $f(n) - n \mid n - 1 \implies f(n) \leq 2n - 1$ , and  $2f(n) - 2n = f(f(n)) - f(n) \mid f(n) - 1 \implies f(n)$  is odd number for all  $n \in \mathbb{N}$ .

Now we prove by induction on  $n$  that  $f(n) = 2n - 1$ . Base case is clear since  $f(1) = 1$ . Now, suppose that  $f(i) = 2i - 1$  for  $1 \leq i < n$ . If  $f(n) = 2i - 1$  for some  $i \neq n$ , it would imply that  $f(n) = f(i)$  or  $f(n) > 2n - 1$ , contradicting the injectivity of  $f$  or  $f \leq 2n - 1$ . Hence,  $f(n) = 2n - 1$  for all  $n \in \mathbb{N}$ .



**Problem - 2.** Let  $\mathbb{R}_{>0}$  denote the set of positive real numbers. Find all pairs  $f, g: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$  of functions such that

$$f(g(x)) = f(x)g(x), \quad f(x) = x(1 + g(x))$$

and such that the sequence  $x, g(x), g(g(x)), \dots$ , takes finitely many different values for all  $x \in \mathbb{R}_{>0}$ .

**Answer:**  $f(x) = x + 1$  and  $g(x) = 1/x$

**Solution:**

The pair above is a solution. In order to prove that there is no other solution, fix  $x \in \mathbb{R}_{>0}$  and denote  $g^0 = x$  and  $g^n = g(g^{n-1})$  for  $n \geq 1$ .

We have  $g(x)(1 + g(g(x))) = f(g(x)) = f(x)g(x) = x(1 + g(x))g(x)$ . Since  $g(x) > 0$ , we have  $1 + g^2 = x(1 + g)$ . This can be rewritten as

$$gg^2 - 1 = (xg - 1)(1 + g).$$

Then  $g^2g^3 - 1 = (gg^2 - 1)(1 + g^2) = (xg - 1)(1 + g)(1 + g^2)$  and more generally

$$g^n g^{n+1} - 1 = (g^{n-1}g^n - 1)(1 + g^n) = (xg - 1)(1 + g)(1 + g^2) \dots (1 + g^n)$$

for any  $n \geq 1$  by induction. There is  $n > m$  with  $g^n = g^m$  by assumption. Then we have

$$(xg - 1)(1 + g) \dots (1 + g^m) \left( (1 + g^{m+1}) \dots (1 + g^n) - 1 \right) = 0.$$

Since  $g^k > 0$ , we must have  $xg - 1 = 0$ . It follows that  $g = 1/x$  and  $f = x + 1$ .

**Problem - 3.** Let  $\mathbb{R}$  denote the set of all real numbers. Find all pairs of functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  and  $h: \mathbb{R}^2 \rightarrow \mathbb{R}$  such that

$$f(x+y-z)^2 = f(xy) + h(x+y+z, xy+yz+zx)$$

for all  $x, y, z \in \mathbb{R}$ .

**Solution:**

Let us define two sets  $D = \{(x+y+z, xy+yz+zx) \mid x, y, z \in \mathbb{R}\}$  and  $E = \{(a, b) \in \mathbb{R}^2 \mid a^2 \geq 3b\}$ . It is obvious that  $D \subseteq E$ . Now let us show that  $E \subseteq D$ . Indeed, if  $a^2 \geq 3b$  there exists  $c$  such that the polynomial  $P(x) = x^3 - ax^2 + bx - c$  has three real roots. Thus,  $D = E$ .

Since the function  $h$  can take any value on  $\mathbb{R}^2 \setminus D$ , we only need to determine the values of  $h$  on  $D$ .

**Answer:**  $f(x) = c$ ,  $h(a, b) = c^2 - c$  or  $f(x) = x/4$ ,  $h(a, b) = (a^2 - 4b)/16$ ,  $(a, b) \in D$

These functions obviously satisfy the equation, so let us show that there is no other solution.

$(x, y, z) \mapsto (x, z, y)$  implies that

$$f(x-y+z)^2 - f(xz) = h(x+y+z, xy+yz+zx) = f(x+y-z)^2 - f(xy) \quad (*)$$

Now if  $x = 1$ , then  $f(y) - f(z) = f(1-y+z)^2 - f(1+y-z)^2$  and  $(y, z) \mapsto (y-z, 0)$  implies that  $f(y-z) - f(0) = f(1-y+z)^2 - f(1+y-z)^2 = f(y) - f(z)$ . And if  $(y, z) \mapsto (x+y, x)$  then  $f(x+y) = f(x) + f(y) - f(0)$ .

Therefore, the function  $g(x) = f(x) - f(0)$  is a Cauchy's function. If  $z = 0$  in  $(*)$ , then  $f(x+y)^2 - f(x-y)^2 = f(xy) - f(0)$  and  $g(xy) = (g(x+y) - g(x-y))(g(x+y) + g(x-y) + 2f(0)) = 4g(y)(g(x) + f(0))$ . Hence,  $f(0)(g(x) - g(y)) = 0$ .

If the function  $g(x)$  is not constant,  $f(0) = 0$  and  $g(xy) = 4g(x)g(y)$ . Thus,  $g(x)$  is an increasing function, so  $g(x) = ax$  and  $g(1) = 4g(1)^2$ ,  $a = 1/4$  and  $f(x) = x/4$ . It is not difficult to see that  $h(a, b) = (a^2 - 4b)/16$  on  $D$ .

$$f: X \rightarrow Y$$

- $f$  monotonic on some interval of  $x$
- $f$  bounded
- $f$  continuous

Cauchy + any condition = linear

$f$  multiplicative;

$$f(x^2) = f(x)^2 \geq 0$$



**Problem - 4.** Let  $\mathbb{R}$  denote the set of all real numbers. Find all functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that

$$f(f(x) + y) = f(f(x) - y) + 4f(x)y$$

for all  $x, y \in \mathbb{R}$ .

**Answer:**  $f(x) = 0$  or  $f(x) = x^2 + c$ .

**Solution:**

It is easy to check that the above function is a solution, so let us prove that there are no other solutions.

Thus we can assume there exists  $c$  such that  $f(c) \neq 0$ . Let  $P(x, y)$  be the given assertion.  $P\left(c, \frac{x}{4f(c)}\right)$

yields  $x = f\left(f(c) + \frac{x}{4f(c)}\right) - f\left(f(c) - \frac{x}{4f(c)}\right)$  then there exists  $u, v \in \mathbb{R}$  such that  $x = f(u) - f(v)$ .

$P(x, f(x))$  yields

$$f(2f(x)) = f(0) + (2f(x))^2 \quad (*)$$

So there exist  $u, v$  such that  $\frac{x}{2} = f(u) - f(v)$ . Then  $P(u, 2f(v) - f(u))$  yields and use  $(*)$

$$f(2f(u) - 2f(v)) + 4f(u)(2f(v) - f(u)) = f(2f(v)) = f(0) + (2f(v))^2$$

this is equivalent to  $f(2f(u) - 2f(v)) = f(0) + (2f(u) - 2f(v))^2$ . Hence  $f(x) = x^2 + f(0)$  for all  $x \in \mathbb{R}$ , so we're done.

**Problem - 5.** Let  $\mathbb{R}_{>0}$  denote the set of positive real numbers. Find all functions  $f: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$  such that

$$f(x)f(y+f(x)) = f(1+xy)$$

for all  $x, y \in \mathbb{R}_{>0}$ .

**Answer:**  $f(x) = 1$  or  $f(x) = 1/x$

**Solution:**

It is easy to check that these are solutions, so we prove there are no other solutions.

**$f = 1$  if  $f$  is not injective:** Suppose  $a > b > 0$  and  $f(a) = f(b)$ . Then we have

$$f(1+ax) = f(a)f(x+f(a)) = f(b)f(x+f(b)) = f(1+bx)$$

for any  $x \in \mathbb{R}_{>0}$ . Let  $c = a/b > 1$  and  $x = y/b$ , then we get  $f(1+cy) = f(1+y)$ . Taking  $a = 1+cy$ ,  $b = 1+y$ , we get

$$f(1+(1+cy)z) = f(1+(1+y)z)$$

for all  $z \in \mathbb{R}_{>0}$ . Now, for  $1 < t < c$ , let  $y = \frac{t-1}{c-t} > 0$  and  $z = \frac{c-t}{c-1} > 0$ . Then  $(1+cy)z = t$  and  $(1+y)z = 1$ , thus  $f(1+t) = f(2)$ . Moreover, we have  $f(1+c^n x) = f(1+x)$  for any  $n \geq 1$  by induction, thus increasing  $n$ , we see that  $f(x) = f(2)$  is constant for all  $x > 2$ .

Finally, for any  $x \in \mathbb{R}_{>0}$ , let  $y = 2 + 1/x$ , then  $y + f(x) > 2$  and  $1 + xy > 2$ , thus  $f(x) = f(1+xy)/f(y+f(x)) = 1$ .

**$f = 1/x$  if  $f$  is injective:** Let  $x > 1$ . For  $y = \frac{x-1}{x} > 0$ , we have  $1+xy = x$ , thus

$$f(1 - 1/x + f(x)) = 1,$$

since  $f(x) > 0$ . Let  $d = f(2) - 1/2$ . Then  $f$  injective implies that  $f(x) = d + 1/x$ . For  $y = x$ , we have  $x > 1$ ,  $y + f(x) > 1$ ,  $1 + xy > 1$ , hence

$$\left(d + \frac{1}{x}\right) \left(d + \frac{1}{x+d+\frac{1}{x}}\right) = d + \frac{1}{1+x^2}.$$

Simplifying to a polynomial identity and considering the constant term, we see that  $d = 0$ . It follows that  $f(x) = 1/x$  for  $x > 1$ . Now let  $y = f(x)$ . Then  $f(2f(x)) = f(1+xf(x))/f(x)$  and thus

$$f\left(\frac{2}{x}\right) = \frac{x}{1+x\frac{1}{x}} = \frac{x}{2}.$$

Therefore  $f(x) = 1/x$  for  $0 < x < 2$ .